CS315 · ALGORITHMS ANALYSIS & DESIGN

Final Drill Solutions

One drill on Lectures 1 to 5 in the midterm format

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01 Solution Q1: MCQ

#	Answer	Reason
1	**c**	A loop inside a loop → n × n
2	**b**	log n levels in the recursion tree, n work on each
3	**c**	n priority-queue operations, each log n
4	**a**	The element is the first one
5	**b**	Greedy fails on 0-1 but works for fractional

02 Solution Q2: counting operations

i runs from 0 to n − 1, so the loop makes **n** iterations. The for header = 1 + (n + 1) = n + 2.

Line	Worst	Best	Why
c ← 0	1	1	One assignment, once
for i ← 0 to n − 1	n + 2	n + 2	i ← 0 once, the test n + 1 times
if A[i] > 0	2n	2n	Index and comparison, n times
c ← c + 1	2n	0	Addition and assignment. Worst: all positive. Best: none
increment i	2n	2n	Addition and assignment, n times
return c	1	1	
Total	**7n + 4**	**5n + 4**	Both are O(n)

03 Solution Q3: sorting

a) Insertion sort

i	key	Array after the iteration
–	–	4 7 1 8 3 5
2	7	4 7 1 8 3 5
3	1	1 4 7 8 3 5
4	8	1 4 7 8 3 5
5	3	1 3 4 7 8 5
6	5	1 3 4 5 7 8

b) Merge sort

Divide	[6 3 8 1 5 2 7 4]			
	[6 3 8 1]		[5 2 7 4]	
	[6 3]	[8 1]	[5 2]	[7 4]
	[6] [3]	[8] [1]	[5] [2]	[7] [4]
Merge	[3 6]	[1 8]	[2 5]	[4 7]
	[1 3 6 8]		[2 4 5 7]	
	[1 2 3 4 5 6 7 8]			

c) Recursion tree

Levels = log(8) + 1 = 3 + 1 = 4 (levels 0 to 3)

Level 2 = 2^2 = 4 problems, each of size $8 \div 2^2 = 2$

Total work = $6n(\log(n) + 1) = 6 \times 8 \times 4 = 192$

04 Solution Q4: hash table

Key	20	34	45	70	56	13
k mod 11	9	1	1	4	1	2

a) Linear probing

Key	k mod 11	Probes	Cell
20	9	9	**9**
34	1	1	**1**
45	1	1 → 2	**2**
70	4	4	**4**
56	1	1 → 2 → 3	**3**
13	2	2 → 3 → 4 → 5	**5**

0	1	2	3	4	5	6	7	8	9	10
	34	45	56	70	13				20	

b) Double hashing

Cell = (h(k) + j·d(k)) mod 11, with j = 0, 1, 2 ...

Key	k mod 11	7 − (k mod 7)	Probes	Cell
20	9	1	9	**9**
34	1	1	1	**1**
45	1	4	1 → 5	**5**
70	4	7	4	**4**
56	1	7	1 → 8	**8**
13	2	1	2	**2**

0	1	2	3	4	5	6	7	8	9	10
	34	13		70	45			56	20	

c) Separate chaining

Index	Chain
0	ø
1	34 → 45 → 56
2	13
3	ø
4	70
5	ø
6	ø
7	ø
8	ø
9	20
10	ø

05 Solution Q5: LCS

If the two characters match: the diagonal + 1. Otherwise: the larger of the cell above and the

cell to the left.

	∅	A	B	D	C	B
∅	0	0	0	0	0	0
B	0	0	1	1	1	1
C	0	0	1	1	2	2
A	0	1	1	1	2	2
D	0	1	1	2	2	2
B	0	1	2	2	2	3

LCS length = **3**. LCS: ADB, BCB, BDB (any one is correct).

06 Solution Q6: matrix chain

Dimensions: p = (6, 4, 10, 2, 8). Each cell = the smallest

`m[i,k] + m[k+1,j] + $p_{i-1} \cdot p_k \cdot p_j$`.

```
m[A,B] (k = A): 0 + 0 + 6·4·10 = 240
```

```
m[B,C] (k = B): 0 + 0 + 4·10·2 = 80
```

```
m[C,D] (k = C): 0 + 0 + 10·2·8 = 160
```

```
m[A,C] (k = A): 0 + 80 + 6·4·2 = 128
```

```
m[A,C] (k = B): 240 + 0 + 6·10·2 = 360
```

```
m[B,D] (k = B): 0 + 160 + 4·10·8 = 480
```

```
m[B,D] (k = C): 80 + 0 + 4·2·8 = 144
```

```
m[A,D] (k = A): 0 + 144 + 6·4·8 = 336
```

```
m[A,D] (k = B): 240 + 160 + 6·10·8 = 880
```

```
m[A,D] (k = C): 128 + 0 + 6·2·8 = 224
```

	A	B	C	D
A	0	240	128	**224**
B	—	0	80	144
C	—	—	0	160
D	—	—	—	0

Minimum = **224** scalar multiplications. Optimal order: **((A(BC))D)**

07 Solution Q7: fractional knapsack

Item	w	v	v ÷ w
A	5	30	6
C	3	18	6
B	10	40	4
D	8	24	3

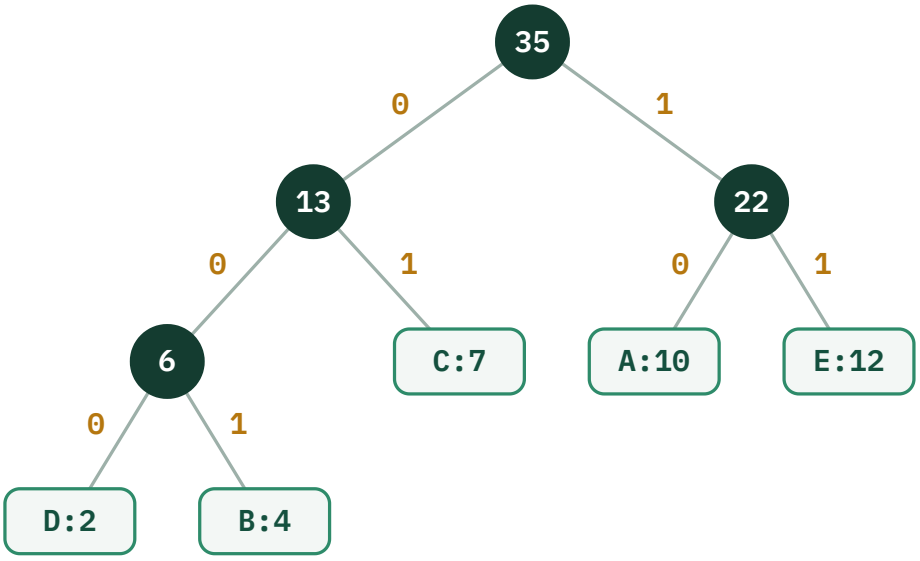
1. Sort by value per kg, largest first.
2. Take all of A: weight 5, value 30.
3. Take all of C: weight 8, value 48.
4. 7 kg are left, so take 7 of the 10 kg of B: `7 ÷ 10 × 40 = 28`.

Total value = 30 + 18 + 28 = **76** (D is not taken).

08 Solution Q8: Huffman

Each time merge the two smallest: D + B = 6, then 6 + C = 13, then A + E = 22, then 13 + 22

= 35.



Char	Freq	Code	Bits
A	10	10	20
B	4	001	12
C	7	01	14
D	2	000	6
E	12	11	24

Total = 20 + 12 + 14 + 6 + 24 = **76 bits** (a fixed 3-bit code needs 35 × 3 = 105).